In [1]:
```
### Training with manually updating W with "Backward" ###

import torch
#from torch.autograd import Variable
import torch.nn as nn
import torch.nn.functional as F

import torch.optim as optim




data = [(1.0,2.1,3.0), (2.0, 3.5, 6.0), (3.0, 3.0, 9.0), (4.0, 2.1, 12.0), (5.0,


class Net(nn.Module):
    def __init__(self):
        super(Net, self).__init__()
        self.fc1 = nn.Linear(2,1,bias=False)  # in dim, out dim
    def forward(self, x):
        x = self.fc1(x)
        return x

net = Net()

print(net)
print(list(net.parameters()))

#input = torch.randn(1)
#out = net(input)

#def criterion(out, label):
#    return (label - out)**2
criterion = nn.MSELoss()


optimizer = optim.SGD(net.parameters(), lr=0.01, momentum=0.5)
#optimizer = optim.Adam(net.parameters(), lr=0.005)

for epoch in range(20): # 0 - 19
    for i, current_data in enumerate(data):
        X1, X2, Y = current_data
        X, Y = torch.FloatTensor([X1, X2]), torch.FloatTensor([Y])
        optimizer.zero_grad()
        outputs = net(X)
        loss = criterion(outputs, Y)
        loss.backward()
        optimizer.step()    ## This line is equivalent to "W = W - lr* W.grad"
        print("Epoch {} - loss: {}".format(epoch, loss))

### Test the trained network ###
for i, current_data in enumerate(data):
    X1, X2, Y = current_data
    X, Y = torch.FloatTensor([X1, X2]), torch.FloatTensor([Y])
    out = net(torch.FloatTensor(X))
    print("when x = {}, y = {}".format(X, out))
```

```
Net(
  (fc1): Linear(in_features=2, out_features=1, bias=False)
)
[Parameter containing:
tensor([[ 0.2841, -0.3596]], requires_grad=True)]
Epoch 0 - loss: 12.047791481018066
Epoch 0 - loss: 36.4962158203125
Epoch 0 - loss: 39.242515563964844
Epoch 0 - loss: 30.495121002197266
Epoch 0 - loss: 21.62595558166504
Epoch 0 - loss: 8.914396286010742
Epoch 1 - loss: 1.3558436632156372
Epoch 1 - loss: 16.855249404907227
Epoch 1 - loss: 28.929401397705078
Epoch 1 - loss: 24.59705924987793
Epoch 1 - loss: 1.7306362390518188
Epoch 1 - loss: 14.119721412658691
Epoch 2 - loss: 0.9993358850479126
Epoch 2 - loss: 12.483170509338379
Epoch 2 - loss: 17.82200813293457
Epoch 2 - loss: 12.068835258483887
Epoch 2 - loss: 0.045652735978364944
Epoch 2 - loss: 9.560882568359375
Epoch 3 - loss: 0.5620837807655334
Epoch 3 - loss: 6.611355781555176
Epoch 3 - loss: 7.961326599121094
Epoch 3 - loss: 4.2429375648498535
Epoch 3 - loss: 0.1064584031701088
Epoch 3 - loss: 4.575018882751465
Epoch 4 - loss: 0.2299918681383133
Epoch 4 - loss: 2.541238307952881
Epoch 4 - loss: 2.5228946208953857
Epoch 4 - loss: 0.9541176557540894
Epoch 4 - loss: 0.25998449325561523
Epoch 4 - loss: 1.5728883743286133
Epoch 5 - loss: 0.06501144170761108
Epoch 5 - loss: 0.6571980714797974
Epoch 5 - loss: 0.4718724191188812
Epoch 5 - loss: 0.07363957166671753
Epoch 5 - loss: 0.24024465680122375
Epoch 5 - loss: 0.34079498052597046
Epoch 6 - loss: 0.009579538367688656
Epoch 6 - loss: 0.07779992371797562
Epoch 6 - loss: 0.01582687348127365
Epoch 6 - loss: 0.012733228504657745
Epoch 6 - loss: 0.14429791271686554
Epoch 6 - loss: 0.022933972999453545
Epoch 7 - loss: 2.1057474441477098e-05
Epoch 7 - loss: 0.0006776386871933937
Epoch 7 - loss: 0.02599821612238884
Epoch 7 - loss: 0.07200641185045242
Epoch 7 - loss: 0.06260064989328384
Epoch 7 - loss: 0.0065550291910767555
Epoch 8 - loss: 0.0015811698976904154
Epoch 8 - loss: 0.02642574906349182
Epoch 8 - loss: 0.06840488314628601
Epoch 8 - loss: 0.08142344653606415
Epoch 8 - loss: 0.019116060808300972
Epoch 8 - loss: 0.029722910374403
Epoch 9 - loss: 0.002724848221987486
```

```
Epoch 9 - loss: 0.03707887977361679
Epoch 9 - loss: 0.06465400010347366
Epoch 9 - loss: 0.05494639277458191
Epoch 9 - loss: 0.003324357559904456
Epoch 9 - loss: 0.03218075633049011
Epoch 10 - loss: 0.002229682635515928 3
Epoch 10 - loss: 0.02782193012535572
Epoch 10 - loss: 0.03937903046607971
Epoch 10 - loss: 0.02635866403579712
Epoch 10 - loss: 6.823960575275123e-05
Epoch 10 - loss: 0.021210839971899986
Epoch 11 - loss: 0.0012364145368337631
Epoch 11 - loss: 0.014498445205390453
Epoch 11 - loss: 0.017300592735409737
Epoch 11 - loss: 0.009089854545891285
Epoch 11 - loss: 0.00026994163636118174
Epoch 11 - loss: 0.009979868307709694
Epoch 12 - loss: 0.0004973221221007407
Epoch 12 - loss: 0.005474641919136047
Epoch 12 - loss: 0.005368749611079693
Epoch 12 - loss: 0.0019799417350441217
Epoch 12 - loss: 0.0006016942788846791
Epoch 12 - loss: 0.003365171840414405
Epoch 13 - loss: 0.00013725647295359522
Epoch 13 - loss: 0.0013785924529656768
Epoch 13 - loss: 0.0009640264324843884
Epoch 13 - loss: 0.0001359023153781891
Epoch 13 - loss: 0.0005400574300438166
Epoch 13 - loss: 0.0007051206193864346
Epoch 14 - loss: 1.9136285118293017e-05
Epoch 14 - loss: 0.00015179517504293472
Epoch 14 - loss: 2.4991608370328322e-05
Epoch 14 - loss: 3.6292920412961394e-05
Epoch 14 - loss: 0.0003184154920745641
Epoch 14 - loss: 4.146341234445572e-05
Epoch 15 - loss: 3.0859723665344063e-09
Epoch 15 - loss: 3.1194795155897737e-06
Epoch 15 - loss: 6.65018887957558e-05
Epoch 15 - loss: 0.00016928557306528091
Epoch 15 - loss: 0.00013581338862422854
Epoch 15 - loss: 1.805886859074235e-05
Epoch 16 - loss: 3.8642169783997815e-06
Epoch 16 - loss: 6.33211457170546e-05
Epoch 16 - loss: 0.0001586144499015063
Epoch 16 - loss: 0.00018437334802001715
Epoch 16 - loss: 4.058383638039231e-05
Epoch 16 - loss: 6.963318446651101e-05
Epoch 17 - loss: 6.249093985388754e-06
Epoch 17 - loss: 8.454528142465279e-05
Epoch 17 - loss: 0.0001454475859645754
Epoch 17 - loss: 0.00012191783753223717
Epoch 17 - loss: 6.7486107582226396e-06
Epoch 17 - loss: 7.278769044205546e-05
Epoch 18 - loss: 4.991741207049927e-06
Epoch 18 - loss: 6.205264071468264e-05
Epoch 18 - loss: 8.699171303305775e-05
Epoch 18 - loss: 5.752572906203568e-05
Epoch 18 - loss: 9.255109034711495e-08
Epoch 18 - loss: 4.704348975792527e-05
Epoch 19 - loss: 2.7188807507627644e-06
```

```
Epoch 19 - loss: 3.178304905304685e-05
Epoch 19 - loss: 3.7567886465694755e-05
Epoch 19 - loss: 1.945465919561684e-05
Epoch 19 - loss: 6.789341568946838e-07
Epoch 19 - loss: 2.1747928258264437e-05
when x = tensor([1.0000, 2.1000]), y = tensor([2.9990], grad_fn=<SqueezeBackward4
>)
when x = tensor([2.0000, 3.5000]), y = tensor([5.9981], grad_fn=<SqueezeBackward4
>)
when x = tensor([3., 3.]), y = tensor([8.9978], grad_fn=<SqueezeBackward4>)
when x = tensor([4.0000, 2.1000]), y = tensor([11.9976], grad_fn=<SqueezeBackward
4>)
when x = tensor([5.0000, 7.2000]), y = tensor([14.9957], grad_fn=<SqueezeBackward
4>)
when x = tensor([ 6.0000, 10.1000]), y = tensor([17.9945], grad_fn=<SqueezeBackwa
rd4>)
```

Q2

The input, target, and weight matrices are:

$$X = [1, 2], \qquad T = [0, 0]$$

$$W_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, W_2 = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

Forward Propagation:

$$S_2 = XW_1 = [1, 2] \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = [2, 3]$$

$$Y_2 = \mathrm{ReLU}(S_2) = [2, 3]$$

$$S_3 = Y_2 W_2 = [2, 3] \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} = [5, 10]$$

$$Y_3 = \mathrm{ReLU}(S_3) = [5, 10]$$

The MSE loss is:

$$L = \frac{(5-0)^2 + (10-0)^2}{2} = 62.5$$

Gradient Calculation for W_2:

$$\frac{\partial L}{\partial Y_3} = Y_3 - T = [5 \quad 10]$$

Since both elements of (S_3=[5,10]) are positive, the ReLU derivative is (1) for both elements. Therefore,

$$\frac{\partial L}{\partial S_3} = [5 \quad 10]$$

$$\frac{\partial L}{\partial W_2} = Y_2^T \frac{\partial L}{\partial S_3}$$

$$\frac{\partial L}{\partial W_2} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} [5 \quad 10] = \boxed{\begin{bmatrix} 10 & 20 \\ 15 & 30 \end{bmatrix}}$$

Gradient Calculation for W_1:

$$\frac{\partial L}{\partial Y_2} = \frac{\partial L}{\partial S_3} W_2^T$$

$$\frac{\partial L}{\partial Y_2} = [5 \quad 10] \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = [25 \quad 25]$$

Since both elements of (S_2=[2,3]) are positive, the ReLU derivative is (1) for both elements. Therefore,

$$\frac{\partial L}{\partial S_2} = [25 \quad 25]$$

$$\frac{\partial L}{\partial W_1} = X^T \frac{\partial L}{\partial S_2}$$

$$\frac{\partial L}{\partial W_1} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [25 \quad 25] = \boxed{\begin{bmatrix} 25 & 25 \\ 50 & 50 \end{bmatrix}}$$

Q4

The network has 3 fully connected layers with the architecture:

$$784 \rightarrow 512 \rightarrow 512 \rightarrow 10$$

The input image is $28 \times 28 = 784$ pixels, and the output layer has 10 neurons.

Total number of parameters:

Layer	Weight parameters	Bias parameters	Total
`fc1`: $784 \rightarrow 512$	$784 \times 512 = 401,408$	512	401,920
`fc2`: $512 \rightarrow 512$	$512 \times 512 = 262,144$	512	262,656
`fc3`: $512 \rightarrow 10$	$512 \times 10 = 5,120$	10	5,130
Total	668,672	1,034	669,706

Answer: 669,706 total parameters.

Total operations per input image:

According to the question:

- 1 MAC = 2 operations
- 1 ReLU activation = 1 operation
- 1 bias addition = 1 operation

Layer	MACs	MAC operations	Bias additions	ReLU operations	Total operations
`fc1`	401,408	802,816	512	512	803,840
`fc2`	262,144	524,288	512	512	525,312
`fc3`	5,120	10,240	10	10	10,260
Total	668,672	1,337,344	1,034	1,034	1,339,412

Answer: 1,339,412 operations per input image.

In [46]:
```
### Training with fancier version ###

import torch
import torch.nn as nn
import torch.nn.functional as F

import torch.optim as optim
import matplotlib.pyplot as plt

class Net(nn.Module): ## nn.Module class is used
    def __init__(self):
        super().__init__()

        self.fc1 = nn.Linear(1, 16, bias=True)
        self.act1 = nn.Tanh()

        self.fc2 = nn.Linear(16, 16, bias=True)
        self.act2 = nn.Tanh()

        self.fc3 = nn.Linear(16, 1, bias=True)

    def forward(self, x):
        x = self.act1(self.fc1(x))
        x = self.act2(self.fc2(x))
        x = self.fc3(x)  # No output activation
        return x

net = Net()

print(net)
print(list(net.parameters())) # parameters are randomized


#def criterion(out, label):
#    return (label - out)**2
criterion = nn.MSELoss()


# optimizer = optim.SGD(net.parameters(), lr=0.1)
optimizer = optim.Adam(net.parameters(), lr=0.01)

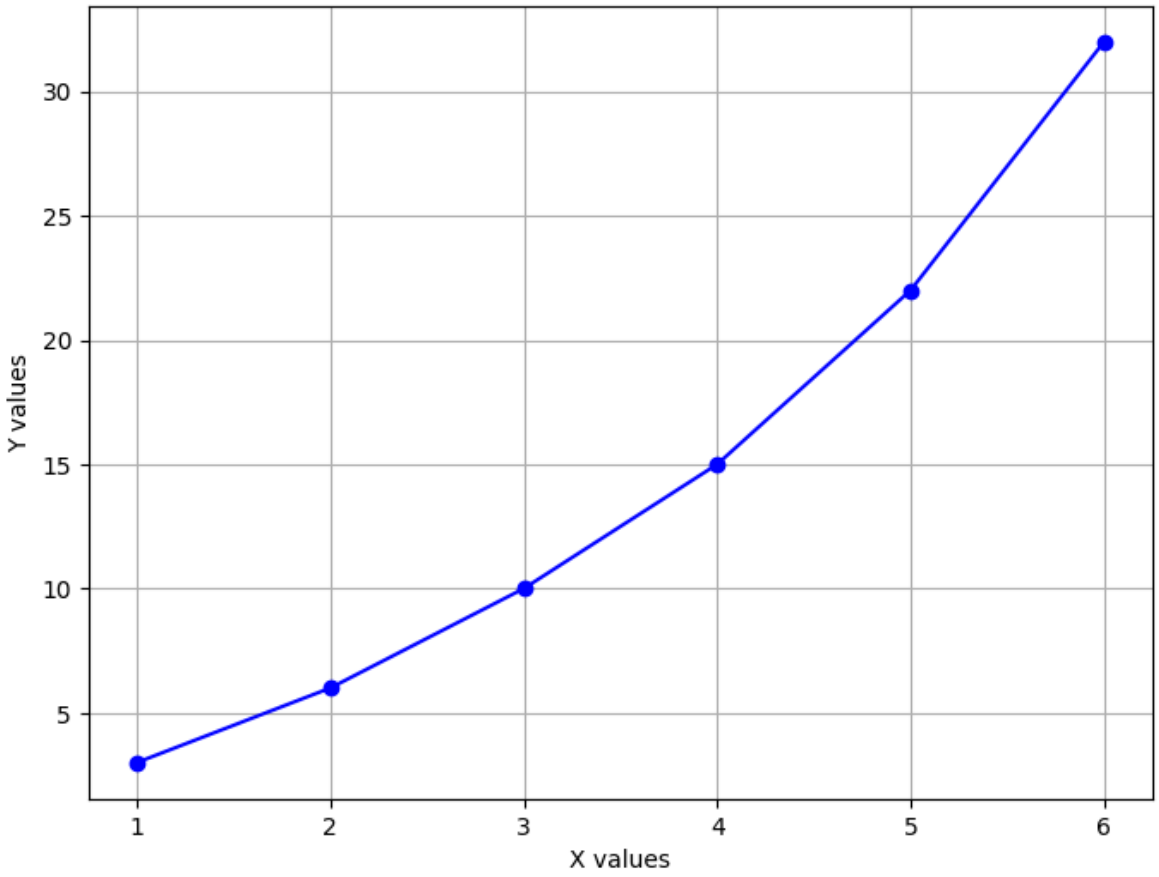

data = [(1.0,3.0), (2.0,6.0), (3.0,10.0), (4.0,15.0), (5.0,22.0), (6.0,32.0)]

# Split data into x and y
x_values, y_values = zip(*data)

# Plotting
plt.figure(figsize=(8, 6))
plt.plot(x_values, y_values, marker='o', linestyle='-', color='b')
plt.xlabel('X values')
plt.ylabel('Y values')
plt.grid(True)
plt.show()
```

```
Net(
  (fc1): Linear(in_features=1, out_features=16, bias=True)
  (act1): Tanh()
  (fc2): Linear(in_features=16, out_features=16, bias=True)
  (act2): Tanh()
  (fc3): Linear(in_features=16, out_features=1, bias=True)
)
[Parameter containing:
tensor([[ 0.3066],
        [-0.9229],
        [-0.8026],
        [ 0.1302],
        [ 0.5766],
        [ 0.2646],
        [ 0.2618],
        [ 0.2404],
        [-0.2466],
        [-0.2804],
        [ 0.1131],
        [ 0.0394],
        [ 0.9090],
        [ 0.5970],
        [ 0.6365],
        [-0.6608]], requires_grad=True), Parameter containing:
tensor([ 0.1325,  0.8033, -0.5506, -0.3546,  0.5854,  0.9352, -0.8683, -0.1927,
         0.8159, -0.0212,  0.1570, -0.6998,  0.8107, -0.0864, -0.9675,  0.2983],
       requires_grad=True), Parameter containing:
tensor([[ 1.2486e-01, -1.1504e-01,  2.1264e-01, -2.8161e-02, -1.1712e-05,
         -2.1211e-01,  1.1329e-01,  7.5698e-02, -1.9750e-01, -4.9676e-02,
         -1.4615e-01,  1.0684e-01, -2.1172e-01,  3.6822e-02,  1.8927e-01,
          8.5638e-02],
        [-1.0363e-01,  8.1149e-02,  3.8339e-02, -1.8632e-01, -8.1954e-02,
         -1.4963e-01,  9.6096e-02,  1.5700e-01, -5.6401e-02,  1.9558e-01,
         -6.8962e-02,  5.7712e-03,  1.3654e-01,  9.6084e-02,  1.4118e-02,
          2.3547e-01],
        [ 2.3134e-01, -5.4165e-02,  1.2739e-01,  5.7898e-02,  6.3178e-02,
         -2.3039e-01, -3.4426e-03,  1.9623e-01,  1.5961e-02, -1.3035e-01,
          1.7061e-01,  4.5013e-02,  2.2824e-01, -1.7025e-01, -4.1148e-03,
          1.3029e-01],
        [-1.9331e-01, -1.6592e-02, -7.2121e-02, -1.4953e-01, -2.4795e-01,
         -8.3776e-02, -2.1978e-01, -1.3275e-01, -1.3984e-01,  2.4766e-01,
          9.4625e-02, -2.1244e-01, -6.7243e-02,  1.2447e-02,  1.8624e-01,
          5.9858e-03],
        [-5.5712e-02, -1.1187e-01,  1.8588e-01,  1.6858e-01,  2.5333e-02,
         -1.7684e-01,  4.2009e-02,  1.2596e-01,  3.4698e-02,  1.9484e-02,
         -1.6755e-01, -6.4004e-02, -4.4138e-02,  1.1440e-01,  1.6970e-01,
         -2.4199e-02],
        [ 1.9406e-01, -1.4151e-01, -1.8894e-01, -1.0455e-01,  8.5770e-02,
         -5.2040e-03, -7.7401e-02,  1.8470e-01,  7.3381e-02,  1.1216e-01,
         -1.4292e-01, -2.2053e-01,  8.7392e-02,  8.0065e-02, -2.4177e-01,
          1.7937e-01],
        [ 1.8826e-01,  2.3717e-01, -1.7893e-01, -1.6329e-01, -1.7726e-01,
          1.0798e-01, -2.3768e-02,  1.3612e-01,  9.9211e-02,  7.6271e-02,
         -2.2183e-01, -8.7248e-02, -9.9108e-02, -8.7263e-02,  2.4252e-01,
          6.2356e-02],
        [-2.3520e-01,  1.9136e-01, -1.4553e-01, -1.6359e-01,  5.4026e-03,
          2.0683e-02,  1.6800e-01, -1.7429e-01, -6.3660e-02, -1.1134e-01,
          9.5414e-02, -1.6232e-01, -2.4632e-01, -2.4110e-01,  1.3283e-01,
          1.6533e-01],
        [-3.6229e-02, -8.2780e-02, -5.3247e-02,  1.9408e-01,  3.1957e-02,
```

```
          -1.4069e-01,  1.2640e-01, -1.2206e-01,  1.2637e-01, -1.5519e-01,
           1.9787e-01,  3.8733e-03, -2.1035e-02,  1.2760e-01,  1.5308e-01,
          -8.8278e-02],
         [ 8.8998e-02, -2.3776e-01, -1.7787e-01,  1.6976e-01,  1.1759e-01,
          -2.5404e-02,  2.1440e-01,  5.4355e-02,  7.0001e-02,  1.9404e-01,
           6.0403e-02, -1.1898e-01, -1.5325e-02, -1.4631e-01,  9.3983e-02,
           3.2773e-02],
         [-2.3197e-01, -1.2320e-01, -1.4360e-01, -1.1840e-01, -1.9032e-01,
          -2.4784e-01, -8.4773e-02, -1.0816e-01, -4.2031e-02, -1.4870e-01,
          -2.3574e-01,  4.4308e-02, -1.6282e-01, -9.5886e-02,  8.2362e-02,
           9.9168e-02],
         [-8.9462e-02,  1.5308e-01,  1.9630e-02, -2.4289e-01, -2.7170e-02,
           2.1260e-01,  1.1191e-01,  1.6689e-01,  1.5487e-01,  4.5005e-02,
           1.9168e-01, -1.6630e-01,  8.7164e-02, -1.4941e-01, -1.5115e-01,
           1.8989e-01],
         [-2.3679e-01,  2.4536e-04,  2.2565e-01,  1.5572e-02, -9.2580e-02,
           5.0499e-02,  1.4292e-01, -2.2007e-02,  1.9103e-01,  1.8569e-01,
          -2.0925e-01, -3.5329e-02, -1.8853e-02,  1.4537e-01,  1.5497e-02,
           1.5717e-01],
         [-2.1134e-01, -2.0270e-01, -1.1706e-01, -6.0780e-02, -2.3009e-01,
          -1.8331e-01,  2.4494e-01, -2.1921e-01, -1.6089e-01,  5.7390e-03,
           1.8717e-02,  1.2163e-01,  1.5871e-01,  1.3625e-01,  1.7880e-01,
           2.0462e-01],
         [ 7.1357e-02, -2.3653e-01, -4.4281e-02, -2.3188e-01, -1.4207e-01,
          -8.7587e-02,  1.5591e-01, -4.0104e-02, -2.2007e-01, -2.2390e-01,
           6.4223e-02,  6.0044e-03, -1.6274e-01,  2.0642e-03, -1.6970e-01,
          -1.5530e-01],
         [ 4.2941e-02,  1.6141e-01, -1.7654e-01,  1.6661e-01, -9.4309e-02,
          -1.6683e-01, -2.1623e-01,  2.1615e-01, -3.4449e-02, -1.6880e-02,
          -2.4114e-01, -7.5681e-03, -1.7238e-01, -1.9718e-01,  2.7711e-02,
          -1.2711e-01]], requires_grad=True), Parameter containing:
tensor([-0.1667, -0.2280,  0.1368, -0.0492, -0.2348, -0.0906,  0.0269,  0.0022,
        -0.1578,  0.1463,  0.0063, -0.2209, -0.1626, -0.2366,  0.2475,  0.1313],
       requires_grad=True), Parameter containing:
tensor([[ 0.1427, -0.1889, -0.1234, -0.0654, -0.1962,  0.1276,  0.1773,  0.2232,
         -0.0556,  0.2215,  0.0580, -0.1891,  0.1508,  0.0231,  0.1620, -0.061
2]],
       requires_grad=True), Parameter containing:
tensor([0.1840], requires_grad=True)]
```



In [47]:
```
for epoch in range(1000):
    for i, current_data in enumerate(data):
        X, Y = current_data
        X, Y = torch.FloatTensor([X]), torch.FloatTensor([Y])
        optimizer.zero_grad()
        outputs = net(X)
        loss = criterion(outputs, Y)
        loss.backward()
        optimizer.step()    ## This line is equivalent to "W = W - lr* W.grad"
    print("Epoch {} - loss: {}".format(epoch, loss))


### Test the trained network ###
for i, current_data in enumerate(data):
    X, Y = current_data
    X, Y = torch.FloatTensor([X]), torch.FloatTensor([Y])
    out = net(torch.FloatTensor(X))
    print("when x = {}, y = {}".format(X, out))
```

```
Epoch 0 - loss: 928.57861328125
Epoch 1 - loss: 858.5050659179688
Epoch 2 - loss: 802.590087890625
Epoch 3 - loss: 754.3414306640625
Epoch 4 - loss: 709.4274291992188
Epoch 5 - loss: 667.6375732421875
Epoch 6 - loss: 629.424072265625
Epoch 7 - loss: 594.7835083007812
Epoch 8 - loss: 563.53173828125
Epoch 9 - loss: 535.4067993164062
Epoch 10 - loss: 510.1085510253906
Epoch 11 - loss: 487.3191223144531
Epoch 12 - loss: 466.70892333984375
Epoch 13 - loss: 447.9522705078125
Epoch 14 - loss: 430.67742919921875
Epoch 15 - loss: 413.66021728515625
Epoch 16 - loss: 394.3923645019531
Epoch 17 - loss: 374.682373046875
Epoch 18 - loss: 356.69561767578125
Epoch 19 - loss: 340.4415283203125
Epoch 20 - loss: 325.405029296875
Epoch 21 - loss: 310.7611083984375
Epoch 22 - loss: 296.2897644042969
Epoch 23 - loss: 282.58563232421875
Epoch 24 - loss: 269.70654296875
Epoch 25 - loss: 257.29998779296875
Epoch 26 - loss: 245.40037536621094
Epoch 27 - loss: 234.19195556640625
Epoch 28 - loss: 223.5659942626953
Epoch 29 - loss: 213.4186248779297
Epoch 30 - loss: 203.80245971679688
Epoch 31 - loss: 194.69949340820312
Epoch 32 - loss: 186.04119873046875
Epoch 33 - loss: 177.82168579101562
Epoch 34 - loss: 170.01943969726562
Epoch 35 - loss: 162.576171875
Epoch 36 - loss: 155.46539306640625
Epoch 37 - loss: 148.66656494140625
Epoch 38 - loss: 142.15110778808594
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Epoch 40 - loss: 129.94515991210938
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Epoch 44 - loss: 108.59790802001953
Epoch 45 - loss: 103.85639953613281
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Epoch 52 - loss: 76.41130065917969
Epoch 53 - loss: 73.21035766601562
Epoch 54 - loss: 70.16889953613281
Epoch 55 - loss: 67.27586364746094
Epoch 56 - loss: 64.5313720703125
Epoch 57 - loss: 61.92073440551758
Epoch 58 - loss: 59.445526123046875
Epoch 59 - loss: 57.08140182495117
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Epoch 62 - loss: 50.52572250366211
Epoch 63 - loss: 48.46856689453125
Epoch 64 - loss: 46.481727600097656
Epoch 65 - loss: 44.549095153808594
Epoch 66 - loss: 42.68949890136719
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Epoch 70 - loss: 35.733642578125
Epoch 71 - loss: 34.1259880065918
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Epoch 73 - loss: 29.768949508666992
Epoch 74 - loss: 31.934091567993164
Epoch 75 - loss: 26.668506622314453
Epoch 76 - loss: 29.49022674560547
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Epoch 79 - loss: 25.869224548339844
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Epoch 84 - loss: 21.63085174560547
Epoch 85 - loss: 18.768627166748047
Epoch 86 - loss: 19.5647029876709
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Epoch 96 - loss: 12.183520317077637
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Epoch 98 - loss: 11.076251029968262
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Epoch 100 - loss: 10.421207427978516
Epoch 101 - loss: 8.091416358947754
Epoch 102 - loss: 9.61473274230957
Epoch 103 - loss: 7.284021377563477
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Epoch 117 - loss: 3.2084133625030518
Epoch 118 - loss: 3.3144404888153076
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when x = tensor([1.]), y = tensor([3.0000], grad_fn=<AddBackward0>)
when x = tensor([2.]), y = tensor([6.0000], grad_fn=<AddBackward0>)
when x = tensor([3.]), y = tensor([10.0000], grad_fn=<AddBackward0>)
when x = tensor([4.]), y = tensor([15.], grad_fn=<AddBackward0>)
when x = tensor([5.]), y = tensor([22.], grad_fn=<AddBackward0>)
when x = tensor([6.]), y = tensor([32.], grad_fn=<AddBackward0>)
```

In [ ]:
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X
```

In [ ]:
```
for epoch in range(20):
    print(epoch)
```

In [ ]:
```
W = torch.tensor([1.0], requires_grad=True)
W = W*2
label = 1.0
loss = W*5 - label
loss.backward()
W.grad
```

In [ ]:

In [ ]:

In [ ]:

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In [25]:
```
import torch
from torch.autograd import Variable
import torch.nn as nn
import torch.nn.functional as F

import torch.optim as optim
import matplotlib.pyplot as plt

class Net(nn.Module):
    def __init__(self):
        super(Net, self).__init__()
        self.fc1 = nn.Linear(2,1,bias=False)
    def forward(self, x):
        x = self.fc1(x)
        return x

net = Net()
print(net)


net.fc1.weight = torch.nn.Parameter(torch.tensor([[1., -1.]], requires_grad=True

print(list(net.parameters()))

#input = torch.randn(1,2)
#out = net(input)


criterion = nn.MSELoss()
#def criterion(out, label):
#    return ((label - out)**2).mean()


optimizer = optim.SGD(net.parameters(), lr=0.01, momentum=0.5)
#optimizer = optim.Adam(net.parameters(), lr=0.005)


data = torch.tensor([[1.,3.], [2.,6.], [3.,9.]], dtype=torch.float)
target = torch.tensor([[1.],[5.],[13.]], dtype=torch.float)

hist = []

############### SGD-based update ###############

for epoch in range(300):
    epoch_loss = 0.0

    for i in range(len(data)):
        optimizer.zero_grad()
        output = net(data[i])
        loss = criterion(output, target[i])
        loss.backward()
        optimizer.step()

        epoch_loss += loss.item()

    # Average loss across all data points for this epoch
    avg_loss = epoch_loss / len(data)
```

```
    hist.append(loss.item())

    print(f"Epoch {epoch} - average loss: {avg_loss:.6f}")

################################################

# ############## Batch GD based update ##############

# for epoch in range(300):
#     optimizer.zero_grad()
#     outputs = net(data)
#     loss = criterion(outputs, target)
#     loss.backward()
#     hist.append(loss.detach())
#     optimizer.step()
#     print("Epoch {} - loss: {}".format(epoch, loss))
# ################################################

### Test the trained network ###
for i, current_data in enumerate(data):
    out = net(current_data)
    print("when x = {}, y = {}".format(current_data, out))

plt.plot(hist, label = "training curve")
```

```
Net(
  (fc1): Linear(in_features=2, out_features=1, bias=False)
)
[Parameter containing:
tensor([[ 1., -1.]], requires_grad=True)]
Epoch 0 - average loss: 39.334532
Epoch 1 - average loss: 80.625629
Epoch 2 - average loss: 37.829956
Epoch 3 - average loss: 75.121512
Epoch 4 - average loss: 37.674733
Epoch 5 - average loss: 74.591525
Epoch 6 - average loss: 37.395082
Epoch 7 - average loss: 74.140797
Epoch 8 - average loss: 37.116333
Epoch 9 - average loss: 73.694979
Epoch 10 - average loss: 36.840476
Epoch 11 - average loss: 73.252864
Epoch 12 - average loss: 36.567522
Epoch 13 - average loss: 72.814407
Epoch 14 - average loss: 36.297446
Epoch 15 - average loss: 72.379595
Epoch 16 - average loss: 36.030199
Epoch 17 - average loss: 71.948314
Epoch 18 - average loss: 35.765771
Epoch 19 - average loss: 71.520611
Epoch 20 - average loss: 35.504124
Epoch 21 - average loss: 71.096446
Epoch 22 - average loss: 35.245239
Epoch 23 - average loss: 70.675761
Epoch 24 - average loss: 34.989076
Epoch 25 - average loss: 70.258529
Epoch 26 - average loss: 34.735609
Epoch 27 - average loss: 69.844735
Epoch 28 - average loss: 34.484837
Epoch 29 - average loss: 69.434354
Epoch 30 - average loss: 34.236708
Epoch 31 - average loss: 69.027334
Epoch 32 - average loss: 33.991201
Epoch 33 - average loss: 68.623627
Epoch 34 - average loss: 33.748307
Epoch 35 - average loss: 68.223287
Epoch 36 - average loss: 33.507994
Epoch 37 - average loss: 67.826226
Epoch 38 - average loss: 33.270226
Epoch 39 - average loss: 67.432434
Epoch 40 - average loss: 33.034979
Epoch 41 - average loss: 67.041878
Epoch 42 - average loss: 32.802236
Epoch 43 - average loss: 66.654501
Epoch 44 - average loss: 32.571973
Epoch 45 - average loss: 66.270310
Epoch 46 - average loss: 32.344158
Epoch 47 - average loss: 65.889284
Epoch 48 - average loss: 32.118780
Epoch 49 - average loss: 65.511377
Epoch 50 - average loss: 31.895798
Epoch 51 - average loss: 65.136550
Epoch 52 - average loss: 31.675208
Epoch 53 - average loss: 64.764818
Epoch 54 - average loss: 31.456959
```

```
Epoch 55 - average loss: 64.396093
Epoch 56 - average loss: 31.241049
Epoch 57 - average loss: 64.030391
Epoch 58 - average loss: 31.027448
Epoch 59 - average loss: 63.667667
Epoch 60 - average loss: 30.816141
Epoch 61 - average loss: 63.307920
Epoch 62 - average loss: 30.607109
Epoch 63 - average loss: 62.951150
Epoch 64 - average loss: 30.400318
Epoch 65 - average loss: 62.597258
Epoch 66 - average loss: 30.195754
Epoch 67 - average loss: 62.246260
Epoch 68 - average loss: 29.993370
Epoch 69 - average loss: 61.898108
Epoch 70 - average loss: 29.793171
Epoch 71 - average loss: 61.552782
Epoch 72 - average loss: 29.595129
Epoch 73 - average loss: 61.210298
Epoch 74 - average loss: 29.399229
Epoch 75 - average loss: 60.870614
Epoch 76 - average loss: 29.205450
Epoch 77 - average loss: 60.533667
Epoch 78 - average loss: 29.013749
Epoch 79 - average loss: 60.199490
Epoch 80 - average loss: 28.824132
Epoch 81 - average loss: 59.868010
Epoch 82 - average loss: 28.636563
Epoch 83 - average loss: 59.539219
Epoch 84 - average loss: 28.451031
Epoch 85 - average loss: 59.213117
Epoch 86 - average loss: 28.267512
Epoch 87 - average loss: 58.889658
Epoch 88 - average loss: 28.085991
Epoch 89 - average loss: 58.568835
Epoch 90 - average loss: 27.906438
Epoch 91 - average loss: 58.250619
Epoch 92 - average loss: 27.728834
Epoch 93 - average loss: 57.934971
Epoch 94 - average loss: 27.553174
Epoch 95 - average loss: 57.621890
Epoch 96 - average loss: 27.379428
Epoch 97 - average loss: 57.311344
Epoch 98 - average loss: 27.207589
Epoch 99 - average loss: 57.003342
Epoch 100 - average loss: 27.037636
Epoch 101 - average loss: 56.697848
Epoch 102 - average loss: 26.869532
Epoch 103 - average loss: 56.394802
Epoch 104 - average loss: 26.703272
Epoch 105 - average loss: 56.094210
Epoch 106 - average loss: 26.538822
Epoch 107 - average loss: 55.796036
Epoch 108 - average loss: 26.376185
Epoch 109 - average loss: 55.500264
Epoch 110 - average loss: 26.215350
Epoch 111 - average loss: 55.206948
Epoch 112 - average loss: 26.056284
Epoch 113 - average loss: 54.915959
Epoch 114 - average loss: 25.898954
```

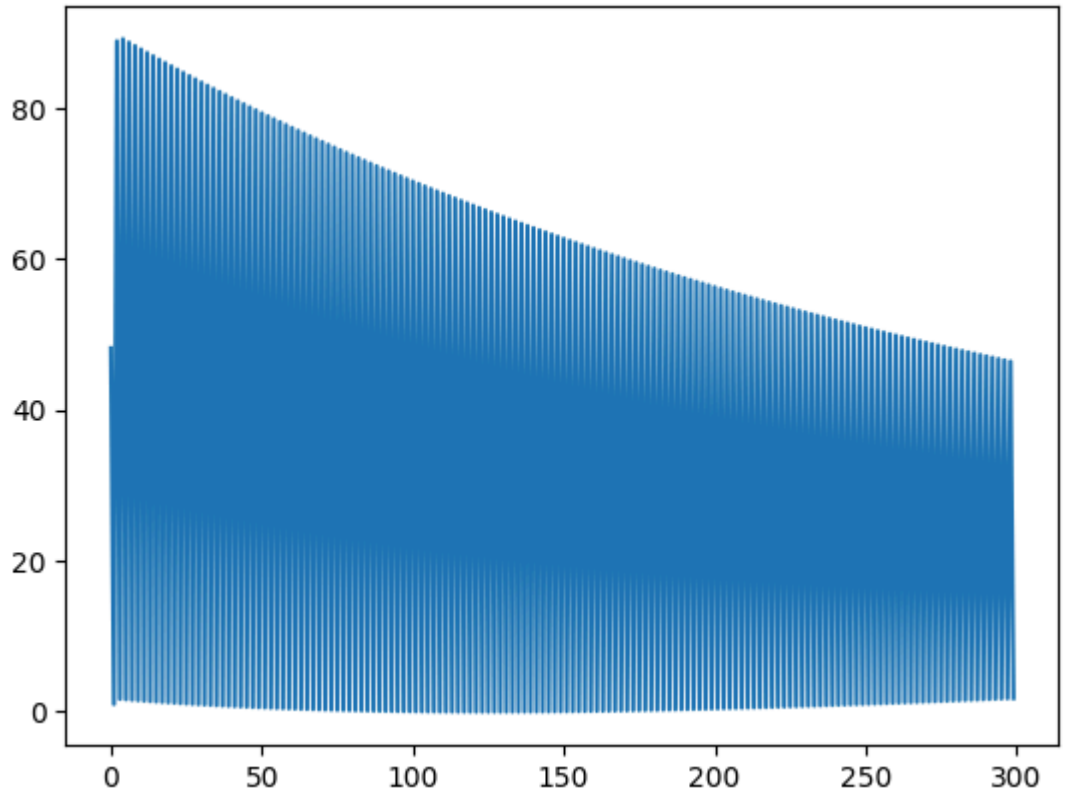
```
Epoch 115 - average loss: 54.627327
Epoch 116 - average loss: 25.743362
Epoch 117 - average loss: 54.341009
Epoch 118 - average loss: 25.589499
Epoch 119 - average loss: 54.057037
Epoch 120 - average loss: 25.437325
Epoch 121 - average loss: 53.775307
Epoch 122 - average loss: 25.286839
Epoch 123 - average loss: 53.495878
Epoch 124 - average loss: 25.138016
Epoch 125 - average loss: 53.218705
Epoch 126 - average loss: 24.990852
Epoch 127 - average loss: 52.943746
Epoch 128 - average loss: 24.845318
Epoch 129 - average loss: 52.671014
Epoch 130 - average loss: 24.701397
Epoch 131 - average loss: 52.400456
Epoch 132 - average loss: 24.559086
Epoch 133 - average loss: 52.132102
Epoch 134 - average loss: 24.418365
Epoch 135 - average loss: 51.865894
Epoch 136 - average loss: 24.279209
Epoch 137 - average loss: 51.601829
Epoch 138 - average loss: 24.141609
Epoch 139 - average loss: 51.339899
Epoch 140 - average loss: 24.005555
Epoch 141 - average loss: 51.080059
Epoch 142 - average loss: 23.871018
Epoch 143 - average loss: 50.822308
Epoch 144 - average loss: 23.737989
Epoch 145 - average loss: 50.566625
Epoch 146 - average loss: 23.606452
Epoch 147 - average loss: 50.312974
Epoch 148 - average loss: 23.476390
Epoch 149 - average loss: 50.061351
Epoch 150 - average loss: 23.347802
Epoch 151 - average loss: 49.811772
Epoch 152 - average loss: 23.220659
Epoch 153 - average loss: 49.564167
Epoch 154 - average loss: 23.094952
Epoch 155 - average loss: 49.318559
Epoch 156 - average loss: 22.970670
Epoch 157 - average loss: 49.074920
Epoch 158 - average loss: 22.847788
Epoch 159 - average loss: 48.833225
Epoch 160 - average loss: 22.726296
Epoch 161 - average loss: 48.593449
Epoch 162 - average loss: 22.606182
Epoch 163 - average loss: 48.355597
Epoch 164 - average loss: 22.487431
Epoch 165 - average loss: 48.119627
Epoch 166 - average loss: 22.370035
Epoch 167 - average loss: 47.885541
Epoch 168 - average loss: 22.253982
Epoch 169 - average loss: 47.653346
Epoch 170 - average loss: 22.139246
Epoch 171 - average loss: 47.422983
Epoch 172 - average loss: 22.025821
Epoch 173 - average loss: 47.194466
Epoch 174 - average loss: 21.913691
```

```
Epoch 175 - average loss: 46.967754
Epoch 176 - average loss: 21.802855
Epoch 177 - average loss: 46.742870
Epoch 178 - average loss: 21.693286
Epoch 179 - average loss: 46.519762
Epoch 180 - average loss: 21.584973
Epoch 181 - average loss: 46.298429
Epoch 182 - average loss: 21.477907
Epoch 183 - average loss: 46.078832
Epoch 184 - average loss: 21.372067
Epoch 185 - average loss: 45.860989
Epoch 186 - average loss: 21.267454
Epoch 187 - average loss: 45.644884
Epoch 188 - average loss: 21.164046
Epoch 189 - average loss: 45.430490
Epoch 190 - average loss: 21.061848
Epoch 191 - average loss: 45.217805
Epoch 192 - average loss: 20.960827
Epoch 193 - average loss: 45.006809
Epoch 194 - average loss: 20.860980
Epoch 195 - average loss: 44.797483
Epoch 196 - average loss: 20.762297
Epoch 197 - average loss: 44.589817
Epoch 198 - average loss: 20.664761
Epoch 199 - average loss: 44.383792
Epoch 200 - average loss: 20.568358
Epoch 201 - average loss: 44.179384
Epoch 202 - average loss: 20.473092
Epoch 203 - average loss: 43.976608
Epoch 204 - average loss: 20.378932
Epoch 205 - average loss: 43.775449
Epoch 206 - average loss: 20.285874
Epoch 207 - average loss: 43.575842
Epoch 208 - average loss: 20.193910
Epoch 209 - average loss: 43.377831
Epoch 210 - average loss: 20.103030
Epoch 211 - average loss: 43.181381
Epoch 212 - average loss: 20.013218
Epoch 213 - average loss: 42.986481
Epoch 214 - average loss: 19.924468
Epoch 215 - average loss: 42.793105
Epoch 216 - average loss: 19.836757
Epoch 217 - average loss: 42.601256
Epoch 218 - average loss: 19.750090
Epoch 219 - average loss: 42.410924
Epoch 220 - average loss: 19.664450
Epoch 221 - average loss: 42.222079
Epoch 222 - average loss: 19.579827
Epoch 223 - average loss: 42.034744
Epoch 224 - average loss: 19.496207
Epoch 225 - average loss: 41.848846
Epoch 226 - average loss: 19.413591
Epoch 227 - average loss: 41.664447
Epoch 228 - average loss: 19.331955
Epoch 229 - average loss: 41.481476
Epoch 230 - average loss: 19.251295
Epoch 231 - average loss: 41.299936
Epoch 232 - average loss: 19.171603
Epoch 233 - average loss: 41.119833
Epoch 234 - average loss: 19.092871
```

```
Epoch 235 - average loss: 40.941138
Epoch 236 - average loss: 19.015087
Epoch 237 - average loss: 40.763858
Epoch 238 - average loss: 18.938237
Epoch 239 - average loss: 40.587955
Epoch 240 - average loss: 18.862306
Epoch 241 - average loss: 40.413407
Epoch 242 - average loss: 18.787303
Epoch 243 - average loss: 40.240273
Epoch 244 - average loss: 18.713204
Epoch 245 - average loss: 40.068455
Epoch 246 - average loss: 18.640002
Epoch 247 - average loss: 39.897980
Epoch 248 - average loss: 18.567691
Epoch 249 - average loss: 39.728843
Epoch 250 - average loss: 18.496266
Epoch 251 - average loss: 39.561031
Epoch 252 - average loss: 18.425712
Epoch 253 - average loss: 39.394542
Epoch 254 - average loss: 18.356017
Epoch 255 - average loss: 39.229325
Epoch 256 - average loss: 18.287177
Epoch 257 - average loss: 39.065420
Epoch 258 - average loss: 18.219181
Epoch 259 - average loss: 38.902777
Epoch 260 - average loss: 18.152022
Epoch 261 - average loss: 38.741392
Epoch 262 - average loss: 18.085696
Epoch 263 - average loss: 38.581285
Epoch 264 - average loss: 18.020186
Epoch 265 - average loss: 38.422424
Epoch 266 - average loss: 17.955477
Epoch 267 - average loss: 38.264767
Epoch 268 - average loss: 17.891583
Epoch 269 - average loss: 38.108359
Epoch 270 - average loss: 17.828477
Epoch 271 - average loss: 37.953154
Epoch 272 - average loss: 17.766161
Epoch 273 - average loss: 37.799174
Epoch 274 - average loss: 17.704621
Epoch 275 - average loss: 37.646370
Epoch 276 - average loss: 17.643849
Epoch 277 - average loss: 37.494748
Epoch 278 - average loss: 17.583841
Epoch 279 - average loss: 37.344303
Epoch 280 - average loss: 17.524585
Epoch 281 - average loss: 37.195034
Epoch 282 - average loss: 17.466076
Epoch 283 - average loss: 37.046916
Epoch 284 - average loss: 17.408308
Epoch 285 - average loss: 36.899950
Epoch 286 - average loss: 17.351261
Epoch 287 - average loss: 36.754091
Epoch 288 - average loss: 17.294941
Epoch 289 - average loss: 36.609367
Epoch 290 - average loss: 17.239341
Epoch 291 - average loss: 36.465764
Epoch 292 - average loss: 17.184452
Epoch 293 - average loss: 36.323295
Epoch 294 - average loss: 17.130258
```

```
Epoch 295 - average loss: 36.181914
Epoch 296 - average loss: 17.076756
Epoch 297 - average loss: 36.041605
Epoch 298 - average loss: 17.023946
Epoch 299 - average loss: 35.902385
when x = tensor([1., 3.]), y = tensor([3.1382], grad_fn=<SqueezeBackward4>)
when x = tensor([2., 6.]), y = tensor([6.2764], grad_fn=<SqueezeBackward4>)
when x = tensor([3., 9.]), y = tensor([9.4147], grad_fn=<SqueezeBackward4>)
```

Out[25]: [<matplotlib.lines.Line2D at 0x236f978bf70>]



In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

In [ ]:

Q6

With batch gradient descent, the network considers the errors from all three examples before deciding how to adjust its weights. The resulting update reflects the combined gradient.

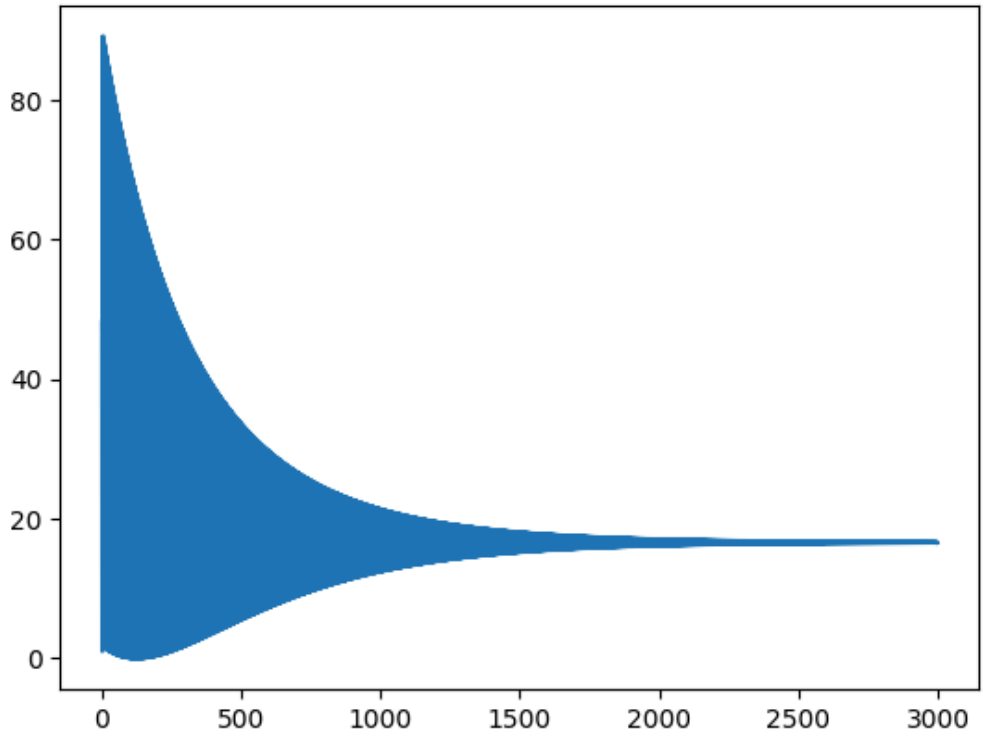
With SGD, the first example changes the weights. The second example then evaluates the model using the already updated weights and may push them in a different direction. The third example may push them in yet another direction.

Consequently, the loss can fluctuate rather than decrease smoothly.

Property	Batch gradient descent	SGD
Gradient calculation	Uses all three samples	Uses one sample at a time
Weight updates per epoch	1	3
Loss curve	Generally smoother	Generally noisier
Gradient estimate	Exact gradient of the full-batch loss	Individual-sample gradient
Computation per update	Higher	Lower
Weight changes during an epoch	Only after all samples	After every sample

Also, after 300 epochs, the loss remains high (envelope: 17.02-35.9). After running for 3000 epochs, the loss envelope decreases (16.01-16.04) but does not reach the low value of 5.47 observed in the Batch Gradient Descent case.

Stochastic gradient descent:



Batch Gradient Descent:

